

MKSSS's Cummins College of Engineering for Women, Pune

Department of Computer Engineering, TY Div B, ICS T1- AY 2025-26

Adversarial Attacks on Machine Learning-based Intrusion Detection Systems (IDS)

Presenters: Harshada Chaudhari, Shruti Desai, Sanskruti Patil

1 Abstract

ML-based Intrusion Detection Systems (IDS) achieve 99%+ accuracy in detecting cyberattacks. However, attackers can fool these systems using Adversarial Machine Learning (AML) – adding tiny invisible changes to attack data so IDS thinks it is normal traffic. This paper surveys all adversarial attack types targeting ML-based IDS, real experimental results, defense strategies, and future research directions.

Key finding: IDS accuracy can drop from 99% to nearly 0% after adversarial attacks.

2 Introduction

Why IDS? Networks face attacks like DoS, DDoS, SQL Injection daily. IDS monitors all traffic and blocks attacks automatically.

Why ML in IDS? Old rule-based IDS missed new attacks. ML learns attack patterns automatically – achieving 99%+ detection accuracy.

The Problem ML models can be fooled. Attackers add tiny changes (δ) to attack traffic – the ML model gets confused and classifies attack as normal. This is an Adversarial Attack.

3 Methods/Materials

Working of ML-based IDS

An ML-based Intrusion Detection System (IDS) analyzes incoming network traffic by extracting features such as packet size, duration, and protocol type. These features are processed by a Machine Learning (ML) model to classify traffic as Normal or Attack.

ML Models Used

- ANN / DNN: Learns complex attack patterns using layered structure
- SVM: Classifies traffic by separating normal and malicious data
- GAN: Generates adversarial samples and supports defense mechanisms

Adversarial Attack Process

Attackers exploit ML models by creating adversarial samples, which are modified versions of attack data with small perturbations (δ).

These changes are minimal but can mislead the model.

Steps:

1. Probe or study the ML model
2. Generate adversarial sample (add δ)
3. Send modified traffic to IDS
4. Model misclassifies → Attack bypasses detection

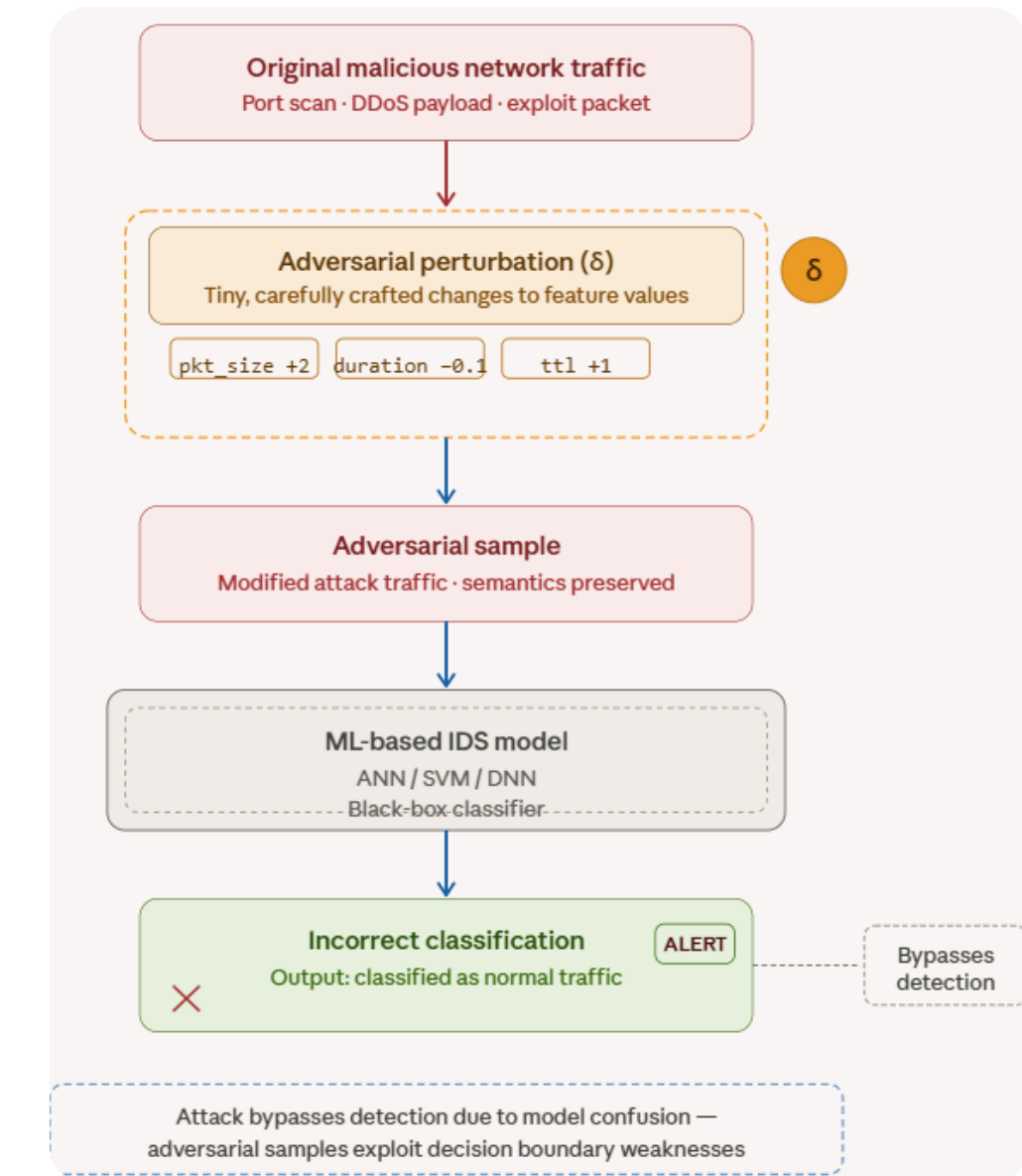
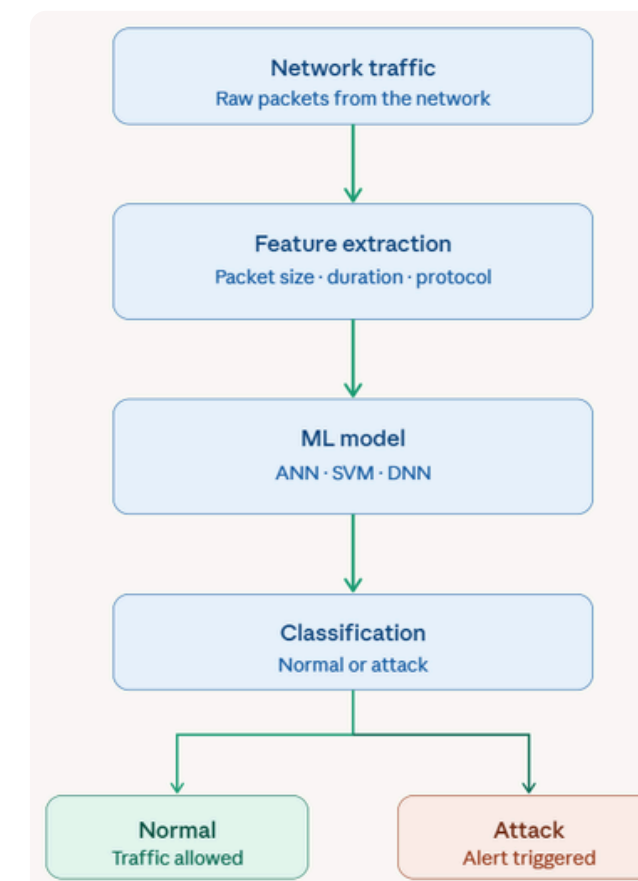
Types of Attacks

- Evasion: Modify input during testing phase
- Poisoning: Inject false data during training
- Extraction: attacker tries to copy or learn the model behavior

Datasets Used

- NSL-KDD – traditional attack dataset
- UNSW-NB15 – modern attack scenarios
- CICIDS2017 – realistic network traffic

Note: Class imbalance affects training and accuracy



5 Discussion

- ML-based IDS rely on learned patterns, making them vulnerable to small input changes that can bypass detection.
- **Black-box (GAN-based)** attacks are especially dangerous as they require no model knowledge and are practical in real-world scenarios.
- Network data is discrete, so many existing defense techniques are less effective compared to other domains.

Limitation & Future Scope:

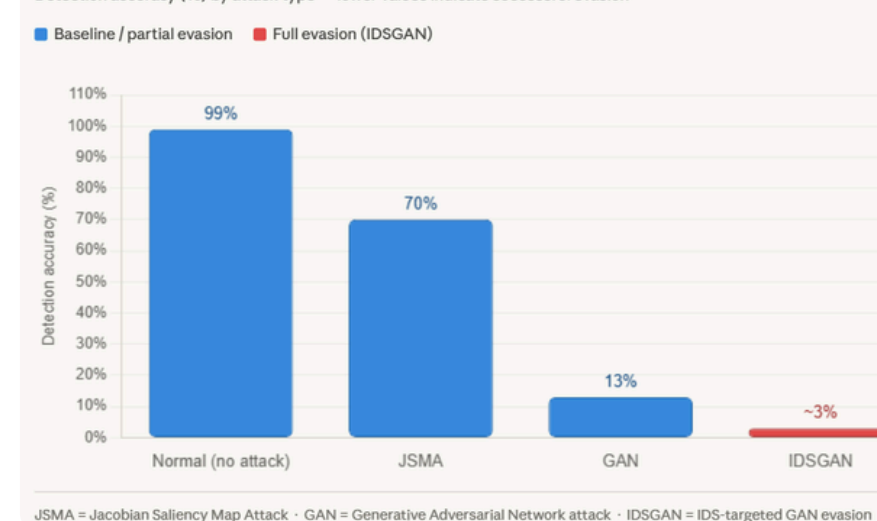
Current defenses are mostly attack-specific and not fully reliable. More robust, adaptable solutions and updated datasets are needed.

4 Results

Experimental studies show that adversarial attacks severely degrade IDS performance. Even highly accurate ML models fail when exposed to carefully crafted inputs.

- Detection accuracy drops significantly under attack
- Black-box attacks are highly effective
- Some models show better resistance than others

Impact of adversarial attacks on IDS accuracy
Detection accuracy (%) by attack type — lower values indicate successful evasion



6 Conclusion

Machine Learning enhances IDS detection but also introduces vulnerabilities to adversarial attacks, making high accuracy alone insufficient for security.

To improve robustness, techniques such as adversarial training, GAN-based detection, and layered security should be applied.

Take-Home Message: IDS must be designed to be both accurate and resilient to attacks for real-world security.

